

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I Levin Anish[192511223] (Name / certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Levin Anish A

Annexure A

Scenario-Based

- A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- Determine the network address and broadcast address for each department.
 - Identify the valid host IP address range for each subnet.
 - Calculate the number of usable host addresses available in each subnet.
 - Determine the remaining unused IP address range within the given network block.
-
- A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- Determine the **network address** and **broadcast address** for each department.
 - Identify the **valid range of host IP addresses**.
 - Calculate the **number of usable hosts** in each subnet.
 - Identify the **unused IP address range** remaining within the allocated IP block.
-
- A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: **2001:0db8:abcd:0000:0000:0000:0000/64**. As the network engineer responsible for the migration, perform the following:

- Convert the given IPv6 address into its **compressed notation**.

- Expand the compressed address back into its **full hexadecimal representation**.
- Identify the **network prefix** using the /64 prefix length.
- Calculate the **total number of host addresses** supported within the subnet.
- the total number of possible host addresses available within this network.

- An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

- The **destination subnet** to which the packet belongs.
- The **matching routing table entry**.
- The **next-hop network/interface** used to forward the packet.
- The **default route** that should be used if the destination does not match any routing table entry.

- A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

- Determine the **network address** and **broadcast address** for each LAN.
- Identify the **valid host IP address range**.
- Suggest suitable IP addresses for **three computers** in each LAN.
- Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

• A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

- Determine the network address and broadcast address for each department.
- Identify the valid host IP address range for each subnet.
- Calculate the number of usable host addresses available in each subnet.
- Determine the remaining unused IP address range within the given network block

Answer : 1

Smart Campus Subnet Design

Given Network: 192.168.100.0/24

Departments:

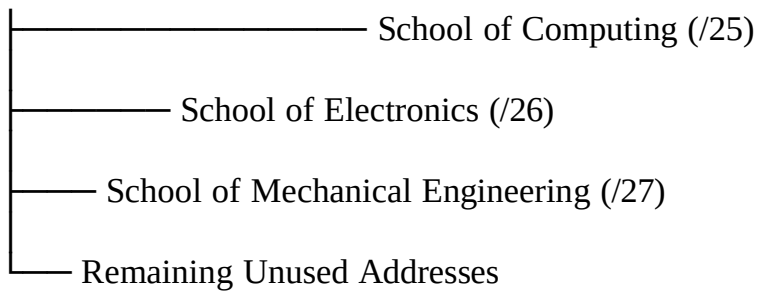
- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

Step 1: Allocate the Subnets

Start allocating from the beginning of the network.

192.168.100.0/24

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Network Diagram

Remaining Unused

192.168.100.224 – 192.168.100.255

Step 2: Subnet Details

Department	Subnet Mask	Network Address	First Host	Last Host	Broadcast Address	Usable Hosts
School of Computing	/25 (255.255.255.128)	192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127	126
School of Electronics	/26 (255.255.255.192)	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191	62
School of Mechanical Engineering	/27 (255.255.255.224)	192.168.10.192	192.168.10.193	192.168.10.222	192.168.10.223	30

Step 3: Remaining Unused IP Address Range

After allocating all three subnets:
Unused Network Range

Start : 192.168.100.224

End : 192.168.100.255

This block contains:

- 32 total IP addresses
- Equivalent subnet: 192.168.100.224/27
- Usable hosts: 30

Address Layout

Final Answer

- Network & Broadcast Address

<i>Department</i>	<i>Network Address</i>	<i>Broadcast Address</i>
<i>School of Computing</i>	<i>192.168.100.0</i>	<i>192.168.100.127</i>
<i>School of Electronics</i>	<i>192.168.100.128</i>	<i>192.168.100.191</i>
<i>School of Mechanical Engineering</i>	<i>192.168.100.192</i>	<i>192.168.100.223</i>

- Valid Host Range

<i>Department</i>	<i>Host Range</i>
<i>School of Computing</i>	<i>192.168.100.1 – 192.168.100.126</i>
<i>School of Electronics</i>	<i>192.168.100.129 – 192.168.100.190</i>
<i>School of Mechanical Engineering</i>	<i>192.168.100.193 – 192.168.100.222</i>

- Usable Hosts

<i>Department</i>	<i>Usable Hosts</i>
<i>School of Computing</i>	<i>126</i>

<i>Department</i>	<i>Usable Hosts</i>
<i>School of Electronics</i>	<i>62</i>
<i>School of Mechanical Engineering</i>	<i>30</i>

- Remaining Unused IP Range

<i>Start Address</i>	<i>End Address</i>	<i>Total Addresses</i>
<i>192.168.100.224</i>	<i>192.168.100.255</i>	<i>32 (30 usable hosts)</i>

This is the complete subnet allocation for the Smart Campus using the 192.168.100.0/24 network.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

- | |
|---|
| 1. Determine the network address and broadcast address for each department. |
| 2. Identify the valid range of host IP addresses . |
| 3. Calculate the number of usable hosts in each subnet. |
| 4. Identify the unused IP address range remaining within the allocated IP block. |

Answer : 2

Given Network

IP Block: 10.10.0.0/16

Department Requirements

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Step 1: Subnet Allocation

Network Diagram

Step 2: Subnet Details

<i>Department</i>	<i>Subnet Mask</i>	<i>Network Address</i>	<i>First Host</i>	<i>Last Host</i>	<i>Broadcast Address</i>	<i>Usable Hosts</i>
<i>Software Development</i>	<i>/24 (255.255.255.0)</i>	<i>10.10.0.0</i>	<i>10.10.0.1</i>	<i>10.10.0.254</i>	<i>10.10.0.255</i>	<i>254</i>
<i>Quality Assurance</i>	<i>/25 (255.255.255.128)</i>	<i>10.10.1.0</i>	<i>10.10.1.1</i>	<i>10.10.1.126</i>	<i>10.10.1.127</i>	<i>126</i>
<i>Human Resources</i>	<i>/26 (255.255.255.192)</i>	<i>10.10.1.128</i>	<i>10.10.1.129</i>	<i>10.10.1.190</i>	<i>10.10.1.191</i>	<i>62</i>
<i>Executive Management</i>	<i>/27 (255.255.255.224)</i>	<i>10.10.1.192</i>	<i>10.10.1.193</i>	<i>10.10.1.222</i>	<i>10.10.1.223</i>	<i>30</i>

Step 3: Remaining Unused IP Address Range

After allocating all four departments, the remaining addresses are:

Start Address	End Address
10.10.1.224	10.10.255.255

Address Layout

Remaining Free IP Addresses

Final Answer

- Network Address & Broadcast Address

<i>Department</i>	<i>Network Address</i>	<i>Broadcast Address</i>
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<i>Software Development</i>	<i>10.10.0.0</i>	<i>10.10.0.255</i>
<i>Quality Assurance</i>	<i>10.10.1.0</i>	<i>10.10.1.127</i>
<i>Human Resources</i>	<i>10.10.1.128</i>	<i>10.10.1.191</i>
<i>Executive Management</i>	<i>10.10.1.192</i>	<i>10.10.1.223</i>

- *Valid Host IP Address Range*

<i>Department</i>	<i>Host Range</i>
<i>Software Development</i>	<i>10.10.0.1 – 10.10.0.254</i>
<i>Quality Assurance</i>	<i>10.10.1.1 – 10.10.1.126</i>
<i>Human Resources</i>	<i>10.10.1.129 – 10.10.1.190</i>
<i>Executive Management</i>	<i>10.10.1.193 – 10.10.1.222</i>

- *Usable Host*

<i>Department</i>	<i>Usable Hosts</i>
<i>Software Development</i>	<i>254</i>
<i>Quality Assurance</i>	<i>126</i>
<i>Human Resources</i>	<i>62</i>
<i>Executive Management</i>	<i>30</i>

- *Remaining Unused IP Address Range*

<i>Start Address</i>	<i>End Address</i>
<i>10.10.1.224</i>	<i>10.10.255.255</i>

This completes the subnet allocation for the company using the 10.10.0.0/16 private IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: **2001:0db8:abcd:0000:0000:0000:0000/64** As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.

2. Expand the compressed address back into its full hexadecimal representation.					
3. Identify the network prefix using the /64 prefix length.					
4. Calculate the total number of host addresses supported within the subnet.					
5. the total number of possible host addresses available within this network.					

Answer : 3

IPv6 Addressing for a Smart Hospital

Given IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000/64

- **Convert the IPv6 Address into Compressed Notation**

Rules:

- Remove leading zeros in each block.
- Replace consecutive groups of 0000 with :: (only once).

Compressed IPv6 Address

2001:db8:abcd::/64

- **Expand the Compressed Address Back to Full Form**

Expand :: back into enough 0000 blocks to make **8 groups**.

Expanded IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000/64

- **Identify the Network Prefix**

(/64) A /64 prefix means:

- First **64 bits** → Network Prefix
- Last **64 bits** → Interface Identifier (Host Portion)

Network Prefix

2001:0db8:abcd:0000::/64

Diagram

IPv6 Address

2001 : 0db8 : abcd : 0000 : 0000 : 0000 : 0000 : 0000

|----- Network Prefix -----| |---- Host-----|

First 64 bits

Last 64 bits

- **Total Number of Host Addresses**

Supported Host bits = $128 - 64 = 64$ bits

Formula:

Number of Hosts = 2^{64}

Calculation:

$2^{64} = 18,446,744,073,709,551,616$

Total Host Addresses:

18,446,744,073,709,551,616

- **Total Number of Possible Host Addresses Available**

Since IPv6 does not reserve separate network and broadcast addresses like IPv4, **all** 2^{64} interface identifiers are available within the subnet.

Total Possible Host Addresses

= 2^{64}

= 18,446,744,073,709,551,616

Network Diagram

Number of Host Bits = 64

Total Hosts = 2^{64}

= 18,446,744,073,709,551,616

Final Answer

Question	Answer
1. Compressed IPv6 Address	2001:db8:abcd::/64
2. Expanded IPv6 Address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
3. Network Prefix (/64)	2001:0db8:abcd:0000::/64
4. Total Host Addresses Supported	$2^{64} = 18,446,744,073,709,551,616$
5. Total Possible Host Addresses	$2^{64} = 18,446,744,073,709,551,616$

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

Answer : 4

Routing Table Lookup Given Routing Table

<i>Destination Network</i>	<i>Subnet Mask</i>
192.168.1.0	/24
192.168.2.0	/24
192.168.3.0	/24

Destination IP Address

192.168.2.45

Step 1: Determine the Destination Subnet

Subnet mask /24 = 255.255.255.0

Perform the AND operation:

Destination IP : 192.168.2.45

Subnet Mask : 255.255.255.0

Network Address: 192.168.2.0

Destination Subnet:

192.168.2.0/24

Step 2: Matching Routing Table Entry

Compare the destination network with the routing table.

<i>Routing Entry</i>	<i>Match?</i>
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192.168.1.0/24	+ No
192.168.2.0/24	Yes
192.168.3.0/24	+ No

Matching Routing Table Entry:

192.168.2.0/24

Step 3: Next-Hop Network / Interface

Since the destination belongs to **192.168.2.0/24**, the router forwards the packet through the interface connected to that network.

Next-Hop Network/Interface:

Interface connected to 192.168.2.0/24

Step 4: Default Route

If the destination IP **does not match** any routing table entry, the router forwards the packet using the **default route**.

The default route is:

0.0.0.0/0

This route sends the packet to the **default gateway (next-hop router)**.

Final Answer

Question	Answer
1. Destination Subnet	192.168.2.0/24
2. Matching Routing Table Entry	192.168.2.0/24
3. Next-Hop Network/Interface	Interface (or next-hop router) connected to 192.168.2.0/24
4. Default Route (if no match)	0.0.0.0/0 (Default Gateway)

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each LAN.
2. Identify the valid host IP address range .
3. Suggest suitable IP addresses for three computers in each LAN.
4. Identify the default gateway that hosts in each LAN should use for communication.

Answer : 5

Network Configuration for Two Office Departments

Given Information

- **Administration LAN:** 192.168.10.0/26
- **Finance LAN:** 192.168.10.64/26

Router Interfaces

- **Administration Gateway:** 192.168.10.1
- **Finance Gateway:** 192.168.10.65

• **Network Address and Broadcast**

Address A /26 subnet has:

- Total IP addresses = **64**
- Usable host addresses = **62**

<i>LAN</i>	<i>Network Address</i>	<i>Broadcast Address</i>
<i>Administration</i>	<i>192.168.10.0</i>	<i>192.168.10.63</i>
<i>Finance</i>	<i>192.168.10.64</i>	<i>192.168.10.127</i>

• **Valid Host IP Address Range**

LAN	First Host	Last Host
Administration	192.168.10.1	192.168.10.62
Finance	192.168.10.65	192.168.10.126

• **Suitable IP Addresses for Three Computers Administration LAN**

Computer	IP Address
PC-1	192.168.10.2
PC-2	192.168.10.3
PC-3	192.168.10.4

Gateway: 192.168.10.1

Finance LAN

Computer	IP Address
PC-1	192.168.10.66
PC-2	192.168.10.67
PC-3	192.168.10.68

Gateway: 192.168.10.65

- Default Gateway for Each LAN

LAN	Default Gateway
Administration	192.168.10.1
Finance	192.168.10.65

Address Layout Diagram

Final Answer

Task	Administration LAN	Finance LAN
Network Address	192.168.10.0	192.168.10.64
Broadcast Address	192.168.10.63	192.168.10.127
Valid Host Range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Example PC IPs	192.168.10.2, 192.168.10.3, 192.168.10.4	192.168.10.66, 192.168.10.67, 192.168.10.68